Linear Diffuser With Nozzles BD1/BD2/BDV



BDV

Description
Aluminum linear diffuser with individually maneuvarable nozzles. Recommended for use in traffic areas, classrooms, laboratories, etc.

Application
Airflow supply.

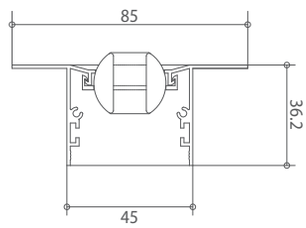
Characteristics
Recommended for rooms between 2.40 and 4.00 m in height.
Long run diffusers are made of assembled modules (maximum 3 m full length). In this case the product is delivered with joining elements between modules.
The spherical design of the nozzles enables air flow redirection in the desired areas.
The spherical nozzles are available in two colors: white RAL 9016 and black RAL 9005.
Depending on the technical and architectural requirements, the diffuser is made with a single row of nozzles - **BD1** or two parallel rows of nozzles - **BD2/BDV**.

Accessories
Attachment spring **A**
Insulated/Non-insulated adapter **ADP IZ/N**

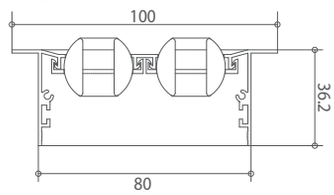
Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural/black anodised aluminum.
Polypropylene nozzles.

Technical design

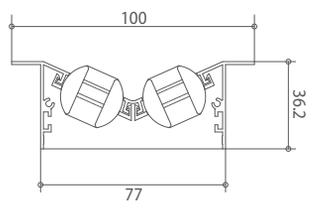
BD1



BD2



BDV



Assembly Methods
Attachment spring.

Selection table BD1, BD2, BDV

Air flow m³/h	Model	BD1		BD2		BDV
		1	2	1	2	3
	Ak(m²)	0.0034	0.0028	0.0067	0.0057	0.0063
30	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	- <20 11.0 2.45	- <20 13.0 2.98			
40	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	1.5 21.0 19.0 3.27	1.5 25.0 23.0 3.97			
50	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	1.6 26.0 29.0 4.08	1.6 30.0 36.0 4.96	1.7 <20 9.0 2.07	1.6 <20 10.0 2.44	- <20 9.0 2.20
60	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	1.8 31.0 41.0 4.90	1.7 35.0 52.0 5.95	1.9 <20 12.0 2.49	1.7 21.0 14.0 2.92	1.8 <20 13.0 2.65
70	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	2.0 35.0 55.0 5.72	1.9 39.0 71.0 6.94	2.1 23.0 16.0 2.90	1.8 25.0 19.0 3.41	1.9 23.0 17.0 3.09
80	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	2.3 38.0 71.0 6.54	2.1 42.0 93.0 7.94	2.3 26.0 21.0 3.32	1.9 29.0 25.0 3.90	2.0 26.0 22.0 3.53
90	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	2.6 41.0 90.0 7.35	2.3 45.0 118.0 8.93	2.5 29.0 27.0 3.73	2.1 32.0 32.0 4.39	2.2 29.0 28.0 3.97
100	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]			2.7 32.0 33.0 4.15	2.2 34.0 39.0 4.87	2.4 31.0 34.0 4.41
110	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]			3.0 34.0 40.0 4.56	2.4 37.0 48.0 5.36	2.6 33.0 41.0 4.85
120	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]			3.2 36.0 47.0 4.98	2.6 39.0 57.0 5.85	2.9 35.0 49.0 5.29

Note: The values of the selection table are given for a length of 1m.

Annotation

- Ak [m²] - Free surface of the diffuser
- X [m] - Length airflow at speed Vk [m/s] = 0.3 m/s
- NR [dB(A)] - Noise level without the room attenuation
- ΔPt [pa] - Pressure loss
- Veff[m/s] - Air velocity in diffuser

COLUMN 1 - nozzles directed toward the residence area
COLUMN 2 - nozzles directed 50% toward the residence area and 50% toward the ceiling
COLUMN 3 - nozzles directed toward the ceiling

Order codes

Type	+	A (Attachment spring)	+	ADP IZ/N	/	Dimensions	/	Nozzles color	/	O
BD1 **BD2** **BDV**		No. of springs depending on the size of the diffuser		Insulated/ Non-insulated adapter		**L** (lenght) on request		White RAL 9016 Black RAL 9005		**0** Painted white RAL 9016 **R** Other RAL paints Black/natural anodised